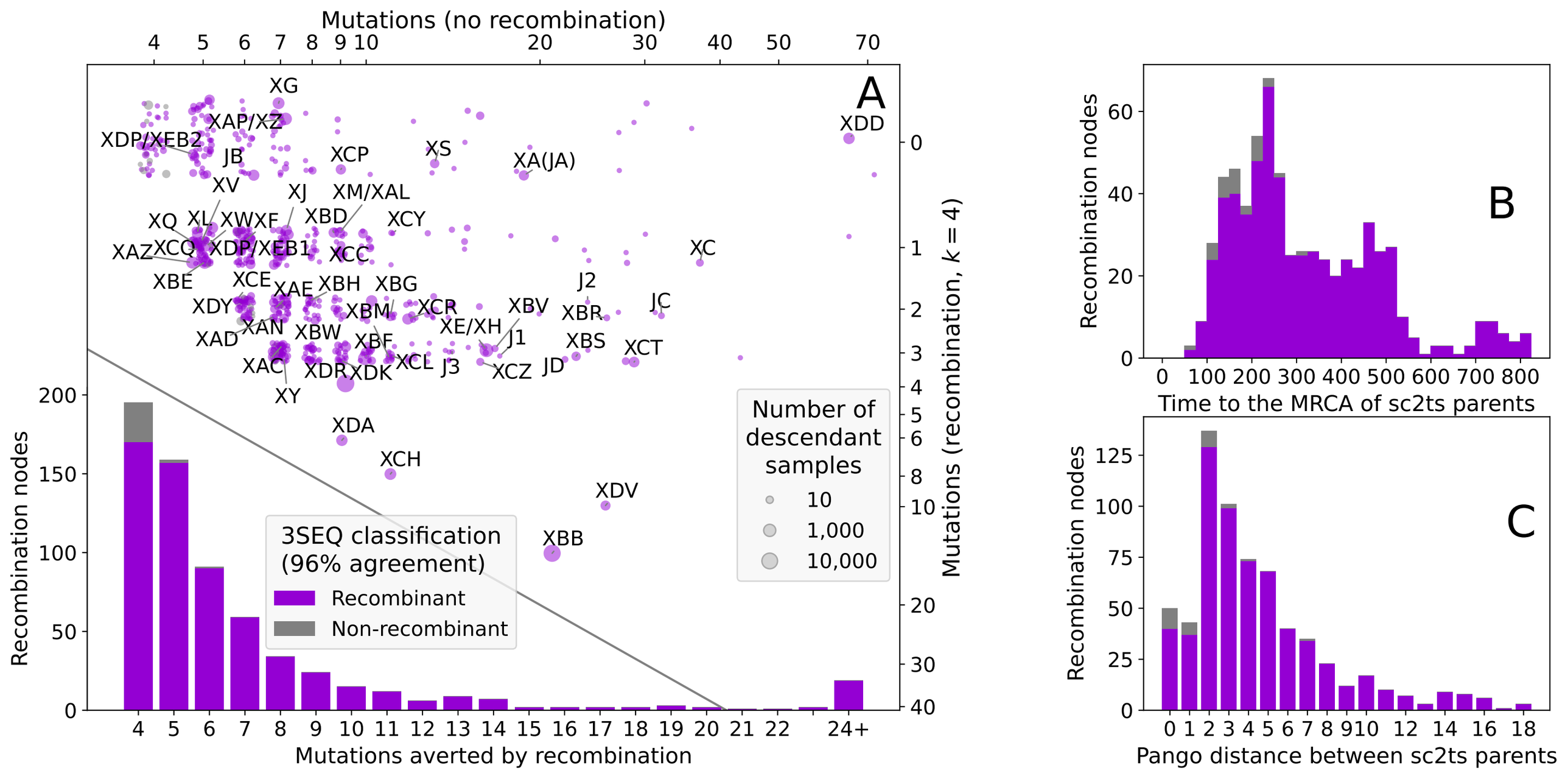

A
Mutations (no recombination)
4 5 6 7 8 9 10 20 30 40 50 70
XG
XAP/XZ
XDP/XEB2
JB
XCP
XS
XA(JA)
XDD
XV
XJ
XM/XAL
XQ
XL
XWXF
XBD
XCY
XAZ
XCQ
XDP/XEB1
XCC
XC
XBE
XCE
XAE
XBH
XBG
J2
XDY
XBM
XCR
XBV
XBR
JC
XAN
XE/XH
XAD
XBW
XBF
J1
XBS
XCT
XAC
XDR
XDK
XCL
J3
XCZ
JD
XY
XDA
XCH
XDV
XBB
Recombination nodes
200
150
100
50
0
Mutations (recombination, k = 4)
0
1
2
3
4
5
6
8
10
20
30
40
Number of descendant samples
10
1,000
10,000
3SEQ classification (96% agreement)
Recombinant
Non-recombinant
4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 24+
Mutations averted by recombination
B
Recombination nodes
60
40
20
0
0 100 200 300 400 500 600 700 800
Time to the MRCA of sc2ts parents
C
Recombination nodes
125
100
75
50
25
0
0 1 2 3 4 5 6 7 8 9 10 12 14 16 18
Pango distance between sc2ts parents